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PREDICTIVE IOT FRAMEWORK FOR EARLY POST-HARVEST SPOILAGE DETECTION

Army Public School, Ahmedabad

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Project Identity

Theme: Science for Sustainable Living

Sub-Theme: Food, Agriculture and Health

Group Leader: Adarsh Mishra

Co-Worker: Ranveer

Guide Teacher: Mrs. Sonika Bohra

District/State: Ahmedabad, Gujarat

OBJECTIVES

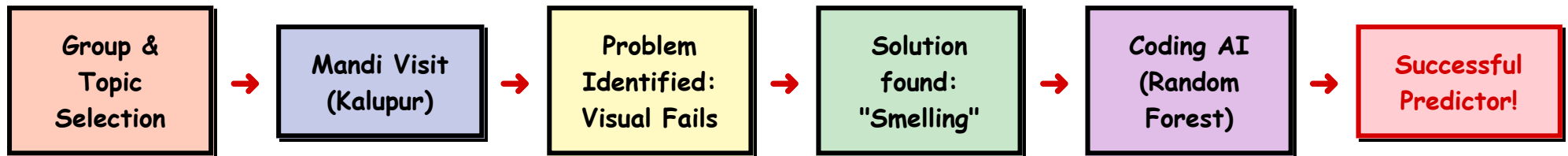
- ✓ To mitigate massive post-harvest crop losses.
- ✓ To detect **Ethanol & VOC gases** from anaerobic fermentation.
- ✓ To intercept spoilage 28 to 36 hours BEFORE visual decay.
- ✓ To build an ultra-low-cost (₹850) AI sensor node.

Paste Photo: You & Ranveer discussing the project

Paste Photo: Large godown cluster near Airport Circle

WORK PLAN

*Horizontal Flow: Left to Right



MAP OF STUDY AREA

Draw/Paste Map of Ahmedabad marking Kalupur, Asarwa, Airport Godowns

Mandis Visited:

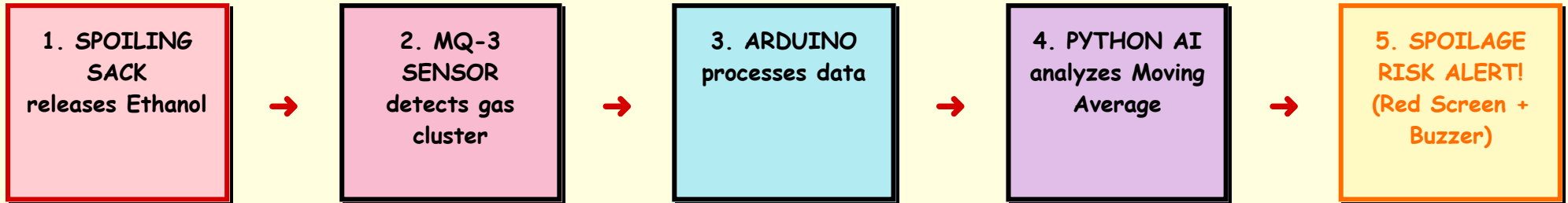
1. Kalupur APMC (Grain)
2. Naroda Road (Potatoes)
3. Asarwa (Cold Storage)

**Observed poor ventilation causes rapid off-gassing.*

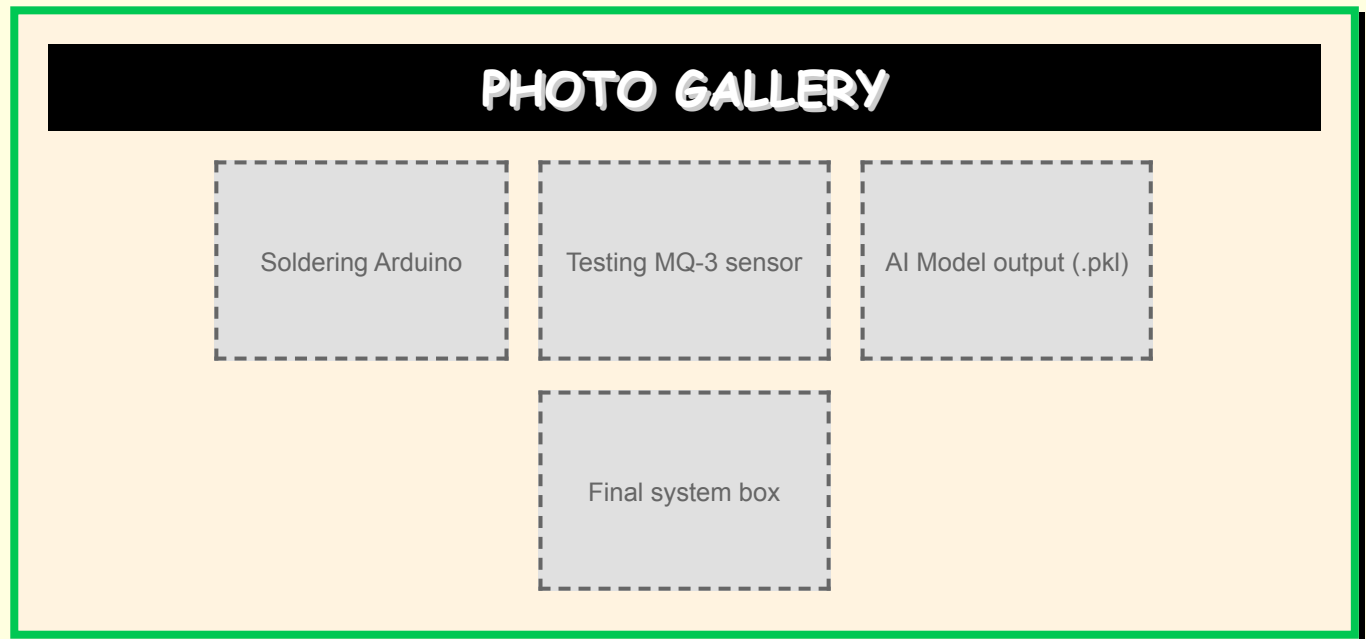
Team at Work

Paste photos of you and Ranveer taking notes in Kalupur

SYSTEM WORKFLOW & AI PIPELINE



The AI Edge: Instead of simple thresholds, we use **Random Forest Regression** to identify long-term gas trends, ignoring short, false spikes (like floor cleaners).



MATERIALS REQUIRED



Arduino Uno



MQ-3 Sensor



Alerts

Total Prototype Cost = ₹850

SCALABILITY MATH

MQ-3 Sensor Radius (indoor aisle) = ~2.2m

**AREA COVERED = $\pi r^2 = 3.14 \times (2.2)^2 =$
~150 sq. ft.**

Standard Sack (1.5 sq ft shelf area)

150 sq ft ÷ 1.5 sq ft = 100 SACKS PER SENSOR!

High ROI: A ₹850 device secures ₹1,00,000+ stock.

CONCLUSION & ULTIMATE SOLUTION

- ✓ Validated: Low-cost sensors can detect invisible chemical spoilage markers.
- ✓ Validated: AI successfully filters false alarms (floor cleaners).
- ✓ **SOLUTION: We give managers a 36-HOUR (1.5 day) Head-Start to isolate rotting sacks before the whole batch ruins, preventing massive waste and protecting food security!**